Executive Briefing

Generative AI at Work

Evidence from 5,172 Customer-Service Agents on the Real Productivity Impact of GenAI

Brynjolfsson, Li & Raymond · NBER WP 31161 · 2024

Agenda

What This Deck Covers

01 Why this study matters

02 Headline findings & trends

03 Core views from the authors

04 Thematic deep dive

05 Strategic insights

06 Recommended actions

07 References

What the data shows

01 Findings & Trends

The Headline Numbers

AI lifts average productivity by 15%, but the gains are deeply uneven.

+15%

Avg. resolutions/hr (RPH) across 5,172 agents

§Abstract; Table 2

+36%

RPH gain for the lowest-skill quintile

§5.2; Table 3

0%

RPH gain for the highest-skill quintile

§5.2; Table 3

−40%

Reduction in new-hire attrition

§7.2

Average gain masks a striking distribution: junior agents leap forward, senior agents barely move. The same intervention, very different outcomes.

AI Flips the Script on Skill-Biased Tech

Earlier IT waves rewarded the strong; GenAI is the first to lift the weak.

Past IT Waves (PC, Internet, ERP)

- Skill-biased technical change
- Top performers benefited most
- Wage inequality widened

Generative AI (this study)

- Skill-equalizing effect
- Lowest-skill agents gain the most
- 2-month new hires match 6-month veterans

Source reference: §Abstract; §5; §6.1

What the authors are arguing

02 Core Views

Thesis: GenAI Is Skill-Equalizing

> Less experienced and lower-skilled workers improve both the speed and quality of their output while the most experienced and highest-skilled workers see small gains in speed and small declines in quality.

— Brynjolfsson, Li & Raymond (2024), §Abstract

Three Mechanisms Behind the Gains

The lift isn't magic — it's encoded best practice flowing from top to bottom.

1. Encoded Tacit Knowledge

AI learns top performers' patterns and broadcasts them to everyone — durable learning, not dependency.

§6.2 outage natural experiment

2. Language Uplift

Conversation analysis shows non-native English speakers' fluency converging toward top-performer style.

§6.3 text analysis

3. Rare-Problem Coverage

Largest gains on rare issues — where human baseline is weakest, AI net contribution is highest.

§6.4 topic stratification

How this changes work itself

03 Themes

Theme

Theme A — The Learning Curve Is Compressed

Ramp-up time collapses; tenure-based seniority loses signal value.

Without AI

- 8–10 months to reach 2.5 RPH
- 25% new-hire attrition within 6 months
- Tenure strongly predicts performance

With AI

- 5 months to reach 3.0 RPH
- ~15% new-hire attrition (−40%)
- Tenure premium weakens

Source reference: §6.1; §7.2

Theme

Theme B — Customer Experience Improves Too

AI assistance doesn't make agents feel robotic — customers respond better.

Customer Behavior

- Fewer aggressive / negative messages
- Manager-escalation requests −25%
- From 6.0% baseline to ~4.5%

Adoption Pattern

- Avg. agent accepts 35% of suggestions
- Adoption rises with exposure
- Skeptical seniors warm up over time

Source reference: §7.1; §6.5

What this means strategically

04 Insights

Strategic Insight: Two Long-Term Risks

Today's productivity win quietly creates two structural problems.

Risk 1 — Top Talent Flattening

If senior agents increasingly defer to AI's median-quality suggestions, their original creative contribution declines — the very signal future models depend on.

Risk 2 — Training Data Decay

Agent conversations are both customer service AND model training data. Without compensation mechanisms, the source of intelligence erodes over time.

Source reference: §8 (Discussion)

What enterprise leaders should do now

05 Recommended Actions

Five Moves for Enterprise Leaders

Treat this study as a playbook, not a forecast.

1

1. Reset HR Economics

Re-evaluate seniority-based pay curves — AI compresses the experience premium.

2

2. Rewire AI ROI Targets

Make new-hire retention a hard KPI for AI deployments, not just speed metrics.

3

3. Treat Conversations as Capital

Establish attribution & incentive systems for employees whose work trains AI models.

4

4. Compress Onboarding

Use AI to shrink customer-service ramp-up from 6–9 months to 2–3 months.

5

5. Protect Top-Talent Originality

Reserve high-difficulty, AI-free tasks for senior staff to preserve creative output.

References & Source

- Brynjolfsson, E., Li, D., & Raymond, L. (2024). Generative AI at Work. NBER Working Paper No. 31161 / arXiv:2304.11771v2 (Nov 7, 2024).
- Published as: Quarterly Journal of Economics, Vol. 140, No. 2, 2025, pp. 889–942.
- Original source: https://arxiv.org/abs/2304.11771

- Sample: 5,172 customer-service agents at a Fortune 500 business-process software firm.
- Identification: Two-way fixed effects DiD; Sun-Abraham robust estimators; outage-based natural experiment for learning persistence.
- All statistics in this deck cited from §Abstract, §5–§8, Tables 2–3, and Figures 2–4 / 10–11 of the paper.

Thank You

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